Test Series Code

12

All India Test Series
NEET (2025-26)

SET

01

DURATION: 180 Minutes | **DROPPER** | **M. MARKS: 720**

DATE: 12/04/2026

SYLLABUS

Physics:	Full Syllabus * As per NTA
Chemistry:	Full Syllabus * As per NTA
Botany:	Full Syllabus * As per NTA
Zoology:	Full Syllabus * As per NTA

GENERAL INSTRUCTIONS:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration and the Test Booklet contains **180** multiple choice questions (four options with a single correct answer).
3. There is only one correct response for each question.
4. Each **correct** answer will give **4 Mark** while **1 Mark** will be **deducted** for a **wrong** answer. The maximum marks are 720.
5. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR INSTRUCTIONS:

1. Use only blue/black dark ballpoint pen.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS): _____

Roll Number: _____

OMR Bar Code Number: _____

Student's Signature: _____ **Invigilator's Signature:** _____

1. The logic gate represented by the following truth table is:

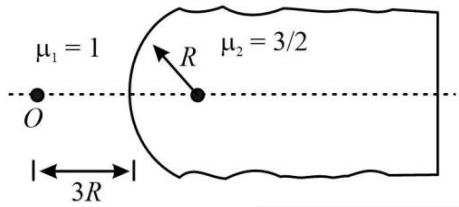
A	B	Y
0	0	0
0	1	0
1	0	1
1	1	0

(1) AND (2) NAND

(3) OR (4) None of these

2. A point object O placed in air lies on the principal axis of a convex spherical surface of radius of curvature R. The surface separates air ($\mu_1 = 1$) from a glass medium ($\mu_2 = 1.5$). The object is placed at a distance of $3R$ from the pole of the surface.

At what distance from the surface, (pole) inside the glass medium, is the real image formed?



(1) $3R$ (2) $6R$

(3) $9R$ (4) $4.5R$

3. A rod with cross-section $A = 2.0\ \text{cm}^2$ is placed in a magnetizing field $H = \dfrac{10^5}{4\pi}\ \text{A m}^{-1}$. The magnetic flux through the rod is $\Phi = 1.0 \times 10^{-5}$ Wb. The magnetic susceptibility χ of the material is (use $\mu_0 = 4\pi \times 10^{-7}\ \text{H m}^{-1}$):

(1) 3 (2) 4

(3) 5 (4) 6

4. The escape velocity from the surface of earth is v_e. If a planet has a mass twice that of earth and a radius half that of earth, the escape velocity from the surface of this planet will be:

(1) v_e (2) $2v_e$

(3) $4v_e$ (4) $\sqrt{2}v_e$

5. A bar magnet oscillates in a uniform magnetic field with a potential energy of U. It is cut perpendicular to its length into two equal parts. These two pieces are then stacked one over another with like poles together. The potential energy of the new system (in the same magnetic field and at the same orientation) is:

(1) $\dfrac{U}{2}$ (2) U

(3) $2U$ (4) $4U$

6. A student performs an experiment with three unknown objects: X, Y, and Z.

Object X repels a glass rod that has been rubbed with silk.

Object Z repels a plastic ruler that has been rubbed with dry hair.

Object Y is attracted to both Object X and Object Z.

What is the nature of Object Y?

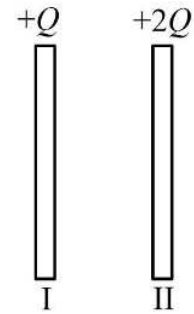
(1) Positively charged

(2) Negatively charged

(3) Electrically Neutral

(4) Could be Positive or Negative

7. Two large, parallel conducting plates, labeled I and II, are given charges of $+Q$ and $+2Q$ respectively. When placed close to each other, what is the charge on the inner surface of Plate I (the surface facing Plate II)?



(1) $+\dfrac{Q}{2}$

(2) $-\dfrac{Q}{2}$

(3) $+\dfrac{3Q}{2}$

(4) $+Q$

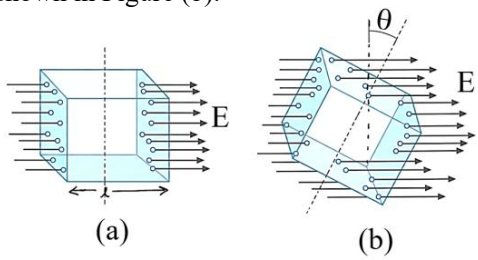
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8. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: When temperature difference across the two sides of a wall is increased, its thermal conductivity increases.
Reason R: Thermal conductivity depends on the nature of material of the wall.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

9. Given below are two statements:
Statement I: The value of the proportionality constant K in Kepler's Third Law ($T^2 = K \cdot R^3$) is the same for the Moon revolving around the Earth and for the Earth revolving around the Sun.
Statement II: The constant K in Kepler's Third Law depends on the mass of the central body around which the object revolves.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

10. The percentage error in the measurement of mass and speed of a body are 2% and 3% respectively. What is the maximum percentage error in the estimation of kinetic energy?
(1) 4% (2) 5%
(3) 8% (4) 1%

11. A cube of edge length l is placed in a region of uniform horizontal electric field of magnitude E. Initially, as shown in Figure (a), one pair of opposite faces of the cube is perpendicular to the electric field lines. The cube is then rotated by an angle θ about a axis perpendicular to the electric field lines, as shown in Figure (b).



What is the total magnitude of the electric flux entering the cube in this new rotated position?
(1) $E \cdot l^2\sqrt{1-\sin(2\theta)}$ (2) $E \cdot l^2\sqrt{1+\cos(2\theta)}$
(3) $E \cdot l^2\sqrt{1+\sin(2\theta)}$ (4) $E \cdot l^2\sqrt{1-\cos(2\theta)}$

12. Two balls, A and B, are projected simultaneously from two points on the same level horizontal ground. Ball A is projected with speed u_1 at an angle θ_1 with the horizontal, and Ball B is projected towards A with speed u_2 at an angle θ_2 with the horizontal. For the balls to collide in mid-air, which of the following conditions must be satisfied?
(1) $u_1\cos\theta_1 = u_2\cos\theta_2$
(2) $u_1\sin\theta_1 = u_2\sin\theta_2$
(3) $u_1 = u_2$
(4) $u_1\sin\theta_2 = u_2\sin\theta_1$

13. The work function of a metal is 4.2 eV . If light of wavelength 2000 Å falls on it, the maximum kinetic energy of the emitted photoelectrons is (take $hc \approx$ 12400 eV Å):
(1) 2.0 eV (2) 4.2 eV
(3) 6.2 eV (4) 1.8 eV

14. The equivalent capacitance between points A and B in a series combination of three capacitors C, $2C$, and $3C$ is:
(1) $6C$ (2) $\frac{6C}{11}$
(3) $\frac{11C}{6}$ (4) $\frac{C}{6}$

15. A force $\vec{F} = \left(3\hat{i} + 4\hat{j}\right)$ N acts on a particle moving from the coordinate $(0,0)$ to $(2\text{m}, 2\text{m})$. The work done by the force is:
(1) 10 J (2) 12 J
(3) 14 J (4) 7 J

16. A wire of resistance R is stretched to twice its original length. This stretched wire is then compressed to $\frac{1}{4}$ of its new length. What is the final resistance of the wire? (the deformations are uniform)
(1) R (2) $4R$
(3) $\frac{R}{4}$ (4) $\frac{R}{16}$

17. The resonant frequency of a series LCR circuit with $L = 2.0$ H, $C = 32\mu F$ and $R = 10\Omega$ is nearest to:
(1) 20 Hz
(2) 125 Hz
(3) 50 Hz
(4) 100 Hz

18. A student is standing in front of "three mirrors stacked one over the other vertically". He observes that his head appears of the same size, the middle part of his body appears smaller, and his legs appear of the same size. What is the sequence of the mirrors from top to bottom? (All images are erect)

(1) Plane, Concave, Plane
(2) Plane, Convex, Plane
(3) Convex, Plane, Convex
(4) Concave, Convex, Plane

19. An electron enters a magnetic field at an angle of 60° to the field lines. The path of the electron is:

(1) Circular (2) Straight line
(3) Parabolic (4) Helical

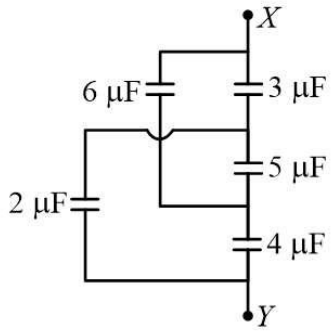
20. Three wires of the same dimensions (length and area) are subjected to the same load. If their elongations are x, $2x$, and $3x$ respectively, the ratio of their Young's modulii is:

(1) $1:2:3$ (2) $3:2:1$
(3) $6:3:2$ (4) $2:3:6$

21. Water flows through a horizontal pipe. At a point where the speed of water is v, the pressure is P. At another point where the speed is $2v$, the pressure will be (density of water = ρ):

(1) $P-\frac{3}{2}\rho v^2$ (2) $P-\rho v^2$
(3) $P-2\rho v^2$ (4) $P+\frac{3}{2}\rho v^2$

22. The equivalent capacitance between points X and Y in figure is:



(1) $\frac{6}{5}\mu F$ (2) $4\mu F$
(3) $\frac{18}{5}\mu F$ (4) $\frac{5}{18}\mu F$

23. If the temperature of a black body is increased by 50%, the power of radiation emitted by it increases by:

(1) 50% (2) 100%
(3) 225% (4) 406%

24. An ideal gas goes through a process $P=\alpha V$, where α is a constant. If the volume increases from V to $2V$, the work done by the gas is:

(1) $\frac{3}{2}\alpha V^2$ (2) αV^2
(3) $2\alpha V^2$ (4) $\frac{1}{2}\alpha V^2$

25. The ratio of the de Broglie wavelength of a proton and an alpha particle accelerated through the same potential difference V is:

(1) $1:2$ (2) $2:1$
(3) $\sqrt{8}:1$ (4) $2\sqrt{2}:1$

26. The efficiency of a Carnot engine is 50% when the sink temperature is 300 K. What should be the source temperature?

(1) 400 K (2) 600 K
(3) 650 K (4) 750 K

27. The rms speed of oxygen molecules at a certain temperature is v. If the temperature is doubled and the oxygen gas dissociates into atomic oxygen, the new rms speed will be:

(1) v (2) $\sqrt{2}v$
(3) $2v$ (4) $4v$

28. A particle executes Simple Harmonic Motion (SHM) with an amplitude of 30 cm. At what distance from the mean position is the potential energy equal to 8 times the kinetic energy?

(1) $10\sqrt{2}$ cm (2) $20\sqrt{2}$ cm
(3) $15\sqrt{2}$ cm (4) $30\sqrt{2}$ cm

29. A circular coil of radius 10 cm has 500 turns and carries a current of 2 A. The magnetic field at the centre of the coil is:

(1) $2\pi\times10^{-3}$ T
(2) $4\pi\times10^{-3}$ T
(3) $2\pi\times10^{-2}$ T
(4) $4\pi\times10^{-2}$ T

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30. In a closed organ pipe, the first overtone frequency is 220 Hz. The length of the pipe is (Speed of sound $=330$ m/s):

(1) 1.125 m
(2) 0.75 m
(3) 1.5 m
(4) 0.375 m

31. In a Young's Double Slit Experiment, the setup is initially illuminated by yellow light. Subsequently, the source is replaced by violet light, keeping all other parameters (D and d) constant.

Given below are two statements:

Statement I: The fringe width decreases when the light source is changed from yellow to violet.

Statement II: The distance between the position of the first minima of yellow light and the first minima of violet light on the same side of the central maximum is equal to half the difference of their respective fringe widths.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

32. The magnetic flux (ϕ) linked with a coil varies with time (t) according to the equation:

$\phi=\left(t^2+3t\right)^2$ milli-webers

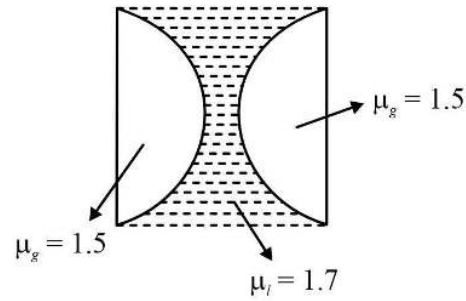
Find the magnitude of the induced EMF at $t=1$ second.

(1) 10 mV
(2) 20 mV
(3) 40 mV
(4) 80 mV

33. The Brewster angle for air to glass interface is i_p. If the refractive index of glass is $\sqrt{3}$, then i_p is:

(1) 30° (2) 45°
(3) 60° (4) 53°

34. An equiconvex lens of focal length f is made of glass with a refractive index $\mu_g=1.5$. It is cut into two thin identical plano-convex lenses by a plane perpendicular to the principal axis. The two halves are then rearranged as shown in figure and the space between them is filled with a thin layer of liquid of refractive index $\mu_l=1.7$. The new equivalent focal length of this combination is:



(1) $5f$ (2) $-5f$
(3) $2.5f$ (4) $-2.5f$

35. In an AC circuit, the voltage is given by $V=100\sin\left(100t\right)$ volts and current by $I=100\sin\left(100t+\pi/3\right)$ A. The power consumed in the circuit is:

(1) 10000 W (2) 3500 W
(3) 5000 W (4) 2500 W

36. For three different materials, the forbidden energy gaps are $E_{g1}=0$ eV , $E_{g2}=1.1$ eV , and $E_{g3}=5.4$ eV. These materials are respectively:

(1) Metal, Insulator, Semiconductor
(2) Semiconductor, Metal, Insulator
(3) Metal, Semiconductor, Insulator
(4) Insulator, Semiconductor, Metal

37. The mass defect in a nuclear reaction is 0.03 u . The energy released in MeV is approximately:

(1) 28 MeV (2) 931 MeV
(3) 0.03 MeV (4) 280 MeV

38. In the hydrogen atom spectrum, which of the following spectral series includes lines that fall within the visible range of the electromagnetic spectrum?

(1) Lyman (2) Balmer
(3) Paschen (4) Brackett

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39. Electromagnetic waves are produced by:

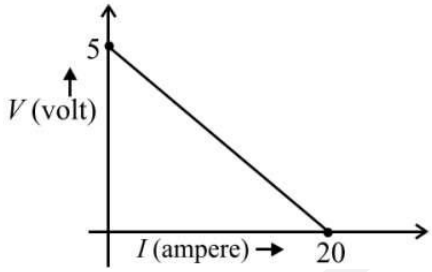
(1) A charge moving at constant velocity

(2) A charge at rest

(3) An accelerated charge

(4) A current-carrying circular coil

40. For a discharging cell, the graph between the potential difference (V) across the terminal of the cells and the current (I) drawn from the cell is shown in figure.

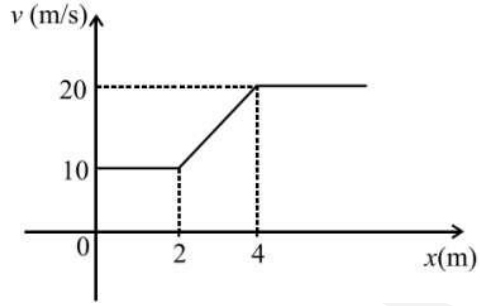


Based upon the above information, the internal resistance of the cell is:

(1) 0.4Ω (2) 2Ω

(3) 0.25Ω (4) 1.3Ω

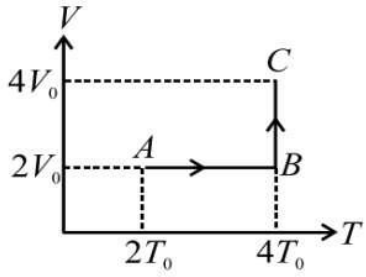
41. For a particle, velocity (v) vs position (x) graph is given. The acceleration of the particle at $x = 3$ m will be:



(1) $20\ \text{m/s}^2$ (2) $30\ \text{m/s}^2$

(3) $75\ \text{m/s}^2$ (4) $60\ \text{m/s}^2$

42. Work done by 2 moles of an ideal gas in process ($A \to B \to C$) shown below is, (where V is volume and T is temperature)



(1) $4RT_0$ (2) $4RT_0\ln(2)$

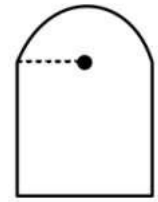
(3) $8RT_0$ (4) $8RT_0\ln(2)$

43. In a resonance tube experiment, first resonating length is 15 cm and second resonating length is 47 cm , then wavelength of the wave is:

(1) 16 cm (2) 32 cm

(3) 64 cm (4) 48 cm

44. One side of a square loop, (length and mass of each side is l and m respectively) is replaced by a semicircular loop of same mass m and radius $\frac{l}{2}$ as shown in figure. The shift in the position centre of mass (from the centre of the square) will be:



(1) $\frac{l}{4\pi}$ (2) $\frac{l}{2\pi}$

(3) $\frac{\pi l}{2}$ (4) $\frac{l}{\pi}$

45. A monkey of mass 40 kg is climbing a vertical massless rope. Match the type of motion described in **List I** with the tension developed in the rope in **List II**. (Take $g = 10\ \text{m/s}^2$)

List-I (Motion of Monkey)		**List-II (Tension in Rope)**	
(A)	Climbing up with acceleration 2 m/s^2	(I)	320 N
(B)	Climbing down with acceleration 2 m/s^2	(II)	400 N
(C)	Climbing up with constant speed 5 m/s	(III)	0 N
(D)	Sliding down freely (frictionless)	(IV)	480 N

Choose the **correct** answer from the options given below:

(1) A-II, B-IV, C-I, D-III

(2) A-IV, B-I, C-II, D-III

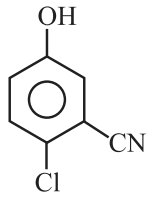
(3) A-IV, B-II, C-I, D-III

(4) A-I, B-IV, C-II, D-III

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46. At certain temperature a 5.12% solution of cane sugar is isotonic with a 0.9% solution of an unknown solute. The molar mass of the solute is:
(1) 60 (2) 46.17
(3) 120 (4) 90

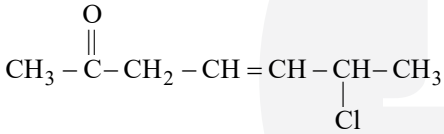
47. The IUPAC name of the following compound is:



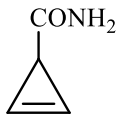
(1) 4-chloro-3-cyanophenol
(2) 2-cyano-4-hydroxychlorobenzene
(3) 6-chloro-3-hydroxybenzonitrile
(4) 2-chloro-5-hydroxybenzonitrile

48. **Correct** order of dipole moment is:
(1) $BF_3 > NF_3 > NH_3$
(2) $NF_3 > BF_3 > NH_3$
(3) $NH_3 > NF_3 > BF_3$
(4) $NH_3 > BF_3 = NF_3$

49. The total number of stereoisomers is:



(1) 4 (2) 6
(3) 8 (4) 12

50. How many products formed in the following reactions (with Br_2 / NaOH , i.e., Hofmann bromamide reaction) can also be prepared by Gabriel phthalimide synthesis?
Given compounds:
1. $Ph-CO-NH_2$
2. $Ph-CH_2-CO-NH_2$
3. $CH_3-CO-NH_2$
4. $CH_3-CO-NH-CH_3$
5. $CH_3-CH_2-CO-NH-CH_3$
6. $C_6H_{11}-CO-NH_2$
7. 

8. $PhCONHCH_3$

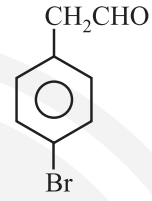
(1) 5 (2) 8
(3) 4 (4) 2

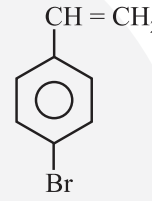
51. Which of the following is most soluble in water?
(1) MnS ($K_{sp} = 8\times10^{-37}$)
(2) ZnS ($K_{sp} = 7\times10^{-16}$)
(3) Bi_2S_3 ($K_{sp} = 1\times10^{-70}$)
(4) Ag_2S ($K_{sp} = 6\times10^{-51}$)

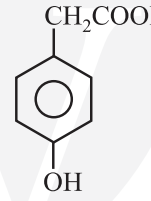
52. What transition in the hydrogen spectrum would have the same wavelength as the Balmer transition n = 4 to n = 2 in the He^+ spectrum?
(1) n = 4 to n = 1
(2) n = 3 to n = 2
(3) n = 3 to n = 1
(4) n = 2 to n = 1

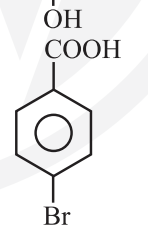
53. $C_6H_6 \xrightarrow[AlCl_3]{C_2H_5Cl} A \xrightarrow[Fe]{Br_2} B \xrightarrow{NBS}$

$C \xrightarrow{aq\ NaOH} D \xrightarrow[\Delta]{\substack{Alkaline\\KMnO_4}} E$

E is:
(1) 

(2) 

(3) 

(4) 

54. 3.011×10^{22} atoms of an element weighs 1.15 g. The atomic mass of the element is :
(1) 23
(2) 10
(3) 16
(4) 35.5

55. $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$
In the reaction given above, the addition of a small amount of inert gas at constant pressure will shift the equilibrium towards which side?
(1) Left hand side
(2) Right hand side
(3) Neither side
(4) Either side

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56. **Correct** order of energy for tetrahedral complex will be as:

(1) $(d_{xy} = d_{yz} = d_{xz}) < (d_{x^2-y^2} = d_{z^2})$

(2) $d_{xy} < (d_{yz} = d_{xz}) < d_{x^2-y^2} < d_{z^2}$

(3) $d_{xy} = d_{yz} < d_{xz}$

(4) $(d_{x^2-y^2} = d_{z^2}) < (d_{xy} = d_{yz} = d_{xz})$

57. Given below are two statements:

Statement I: Fractional oxidation numbers are commonly observed in mixed oxides such as Fe_3O_4.

Statement II: Fe_3O_4 contains both Fe^{2+} and Fe^{3+} ions.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.

(2) Statement I is incorrect but Statement II is correct.

(3) Both Statement I and Statement II are correct.

(4) Both Statement I and Statement II are incorrect.

58. If rate constant for a reaction is 10^{-3} sec^{-1}. Then time for the completion of 75% reaction is:

(1) 693 sec. (2) 1386 sec.

(3) 6930 sec. (4) 2029 sec.

59. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Enantiomers have identical physical properties except for the direction of rotation of plane-polarized light.

Reason R: Enantiomers are non-superimposable mirror images.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true but R is NOT the correct explanation of A.

60. The specific conductivity of 0.1 N KCl solution is 0.0129 Ω^{-1} cm^{-1}. The resistance of the solution in the cell is 100 Ω. The cell constant of the cell will be:

(1) 1.10 (2) 1.29

(3) 0.56 (4) 2.80

61. Given below are two statements:

Statement I: The bond order of O_2^- is lower than that of O_2.

Statement II: Addition of an electron to O_2 occurs in an antibonding orbital.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.

(2) Statement I is incorrect but Statement II is correct.

(3) Both Statement I and Statement II are correct.

(4) Both Statement I and Statement II are incorrect.

62. $CH_3 - CH = CH_2 + NOCl \rightarrow P.$

Identify the adduct

(1) $CH_3 - \underset{\displaystyle Cl}{\underset{|}{CH}} - \underset{\displaystyle NO}{\underset{|}{CH_2}}$

(2) $CH_3 - \underset{\displaystyle NO}{\underset{|}{CH}} - \underset{\displaystyle Cl}{\underset{|}{CH_2}}$

(3) $CH_3CH_2 - \underset{\displaystyle NO}{\underset{|}{CH}} - Cl$

(4) $\underset{\displaystyle NO}{\underset{|}{CH_2}} - CH_2 - \underset{\displaystyle Cl}{\underset{|}{CH_2}}$

63. The enthalpy change for the transition of liquid water to steam is $\Delta H_{vap} = 37.3$ kJ mol^{-1} at 373 K. The entropy change for the process is:

(1) 132.5 J mol^{-1} K^{-1} (2) 100 J mol^{-1} K^{-1}

(3) 135.3 J mol^{-1} K^{-1} (4) 75.5 J mol^{-1} K^{-1}

64. The alkene which gives 4, 4-dimethyl pent-an-2-one on reductive ozonolysis is

(1) $CH_3 - \underset{\displaystyle CH_3}{\underset{|}{\overset{\displaystyle CH_3}{\overset{|}{C}}}} - CH_2 - \underset{\displaystyle CH_3}{\underset{|}{C}} = CH_2$

(2) $CH_3 - \underset{\displaystyle CH_3\ CH_2}{\underset{| \quad ||}{\overset{\displaystyle CH_3}{\overset{|}{C}}}} - C - CH_2CH_3$

(3) $CH_3 - \underset{\displaystyle CH_3}{\underset{|}{\overset{\displaystyle CH_3}{\overset{|}{C}}}} - CH_2 - CH = CH_2$

(4) $CH_3 - \underset{\displaystyle CH_3}{\underset{|}{\overset{\displaystyle CH_2}{\overset{||}{C}}}} - \overset{\displaystyle CH_3}{\overset{|}{C}} - CH_2 - CH_3$

[Contd...

65. The approximate time duration in hours to electroplate 30 g of calcium from molten calcium chloride using a current of 5 A is
[At. mass of Ca = 40]
(1) 80 (2) 10
(3) 16 (4) 8

66. Match **List-I** with **List-II.**

List-I (Orbital)		List-II (Radial Nodes)	
(A)	3s	(I)	2
(B)	3p	(II)	1
(C)	3d	(III)	0
(D)	4s	(IV)	3

Choose the **correct** answer from the options given below:
(1) A-I, B-II, C-III, D-IV
(2) A-IV, B-III, C-II, D-I
(3) A-II, B-III, C-IV, D-I
(4) A-II, B-I, C-III, D-IV

67. For a reaction A $\rightarrow$ B; ΔH = 20 kJ mol^{-1} the activation energy of the forward reaction is 85 kJ/mol. The activation energy of the backward reaction will be
(1) 105 kJ/mol
(2) 65 kJ/mol
(3) 45 kJ/mol
(4) 75 kJ/mol

68. $CH_3 - CH_2 - C \equiv N$ is treated with the following reagents sequentially:
(i) OH^- / H_2O
(ii) H_3O^+
(iii) Red P / Cl_2
(iv) H_2O
The IUPAC name of the major product is:
(1) 2-chloropropanoic acid
(2) 3-chloropropanoic acid
(3) Propanoyl chloride
(4) 2-hydroxypropanoic acid

69. Match **List-I** with **List-II.**

List-I (Condition)		List-II (ΔS for isothermal expansion)	
(A)	$V_2 = 2V_1$	(I)	nR ln 2
(B)	$V_2 = 3V_1$	(II)	nR ln 3
(C)	$V_2 = V_1$	(III)	0
(D)	$V_2 = eV_1$	(IV)	nR

Choose the **correct** answer from the options given below:
(1) A-I, B-II, C-III, D-IV
(2) A-IV, B-III, C-II, D-I
(3) A-II, B-III, C-IV, D-I
(4) A-II, B-I, C-III, D-IV

70. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Atomic orbitals must have comparable energies for effective overlap.
Reason R: Orbitals with very different energies cannot overlap effectively.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

71. Given below are two statements:
Statement I: When aluminium reacts with NaOH, $[Al(OH)_6]^{3-}$ is formed.
Statement II: Shape of H_2O, XeO_4 and ClO_4^- are bent, pyramidal and tetrahedral respectively.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Both Statement I and Statement II are correct
(2) Statement I is correct and Statement II is incorrect
(3) Statement I is incorrect and Statement II is correct
(4) Both Statement I and Statement II are incorrect

72. Match **List-I** with **List-II.**

List-I (Compound)		List-II Van't Hoff factor (assume completely dissociation)	
(A)	NaCl	(I)	2
(B)	$CaCl_2$	(II)	3
(C)	Glucose	(III)	1
(D)	$AlCl_3$	(IV)	4

Choose the **correct** answer from the options given below:
(1) A-I, B-II, C-III, D-IV
(2) A-IV, B-III, C-II, D-I
(3) A-II, B-III, C-IV, D-I
(4) A-II, B-I, C-III, D-IV

[Contd...

73. In the reaction, A → Products, when the concentration of A was reduced from 2.4×10^{-2} M to 1.2×10^{-2} M, the rate decreased 8 times at the same temperature. The order of the reaction is

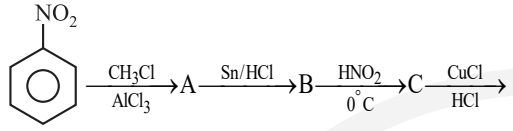
(1) 0 (2) 1

(3) 2 (4) 3

74. The reagents required to prepare m-chloro toluene are:

(1) $\text{Toluene} \xrightarrow[\text{(2) sunlight}]{\text{(1) } Cl_2 \text{ excess}} A \xrightarrow[AlCl_3]{Cl_2} B \xrightarrow{Zn,\ HCl}$

(2) $\text{Toluene} \xrightarrow[AlCl_3]{Cl_2} A \xrightarrow[AlCl_3]{Cl_2}$

(3) 

(4) $\text{Chlorobenzene} \rightarrow \xrightarrow[AlCl_3]{CH_3Cl}$

75. Which one of the following statements is **not** true for glucose?

(1) α-D (+)-glucose undergoes mutarotation

(2) It has four asymmetric carbons is Fischer projection formula

(3) It gives saccharic acid with Tollen's reagent

(4) It reacts with hydroxyl amine

76. At 400 K, in a 1.0 L vessel, N_2O_4 is allowed to attain equilibrium, $N_2O_4(g) \rightleftharpoons 2NO_2(g)$. At equilibrium, the total pressure is 600 mm Hg, when 20% of N_2O_4 is dissociated. The value of K_p for the reaction is:

(1) 50

(2) 100

(3) 150

(4) 200

77. Match **List-I** with **List-II.**

List-I (Species)		List-II (Bond order)	
(A)	O_2	(I)	1
(B)	O_2^-	(II)	2
(C)	N_2^-	(III)	1.5
(D)	B_2	(IV)	2.5

Choose the **correct** answer from the options given below:

(1) A-I, B-II, C-III, D-IV

(2) A-IV, B-III, C-II, D-I

(3) A-II, B-III, C-IV, D-I

(4) A-II, B-I, C-III, D-IV

78. Given below are two statements:

Statement I: In an adiabatic process, heat exchange with surroundings is zero.

Statement II: Temperature of the system always remains constant in an adiabatic process.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.

(2) Statement I is incorrect but Statement II is correct.

(3) Both Statement I and Statement II are correct.

(4) Both Statement I and Statement II are incorrect.

79. Which of the following is pH of solution formed by mixing 0.2 M NH_4Cl and 0.1 M NH_3? The pk_b of ammonia solution is 4.75.

(1) 9.85 (2) 7.05

(3) 6.95 (4) 8.95

80. Which of the statements about "Denaturation" given below are **correct**?

A. Denaturation of proteins causes loss of secondary and tertiary structures of the protein.

B. Denaturation leads to the conversion of double strand of DNA into single strand.

C. Denaturation affects primary structure which gets distorted.

(1) B and C

(2) A and C

(3) A and B

(4) A, B and C

81. Two particle A and B are in motion. If the wavelength associated with 'A' is 33.33 nm , the wavelength associated with 'B' whose momentum is $\frac{1}{3}$rd of 'A' is:

(1) 1.0×10^{-8} m

(2) 2.5×10^{-8} m

(3) 1.25×10^{-7} m

(4) 1.0×10^{-7} m

82. The molal elevation constant of water is 0.52 °C. The boiling point of 1.0 molal aqueous KCl solution (assuming complete dissolution of KCl) therefore should be:

(1) 98.96 °C

(2) 100.52 °C

(3) 101.04 °C

(4) 107.0 °C

[Contd...

83. Given below are two statements:

Statement I: The amount of substance deposited during electrolysis is proportional to the quantity of electricity passed.

Statement II: One faraday of electricity deposits one gram of any substance.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.

(2) Statement I is incorrect but Statement II is correct.

(3) Both Statement I and Statement II are correct.

(4) Both Statement I and Statement II are incorrect.

84. Which one of the following order is **correct** for the first ionization energies of the elements?

(1) B < Be < N < O

(2) Be < B < N < O

(3) B < Be < O < N

(4) B < O < Be < N

85. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Order of a reaction is always equal to the stoichiometric coefficient of reactants.

Reason R: Order depends on the mechanism of the reaction.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true but R is NOT the correct explanation of A.

86. DDT can be synthesized from the chlorination product of which one of the following aldehydes?

(1) HCHO

(2) H_3CCHO

(3) H_3CCH_2CHO

(4) C_6H_5CHO

87. In which of the following species, the central atom has the maximum number of lone pair of electrons?

(1) SF_4

(2) XeF_4

(3) ClO_3^-

(4) I_3^-

88. (A) 0.4 moles of cyclohexane

(B) 68.4 g of sucrose

(C) 70.2 L of oxygen at STP

The order of total number of atoms present in the above samples:

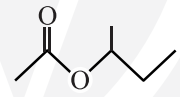
(1) C > A > B

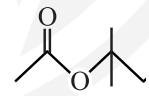
(2) B > A > C

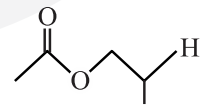
(3) A > B > C

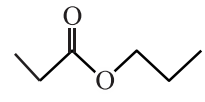
(4) C > B > A

89. An organic compound (A) (molecular formula $C_6H_{12}O_2$) was hydrolysed with dil. H_2SO_4 to give a carboxylic acid (B) and an alcohol (C). 'C' gives white turbidity immediately when treated with anhydrous $ZnCl_2$ and conc. HCl. The organic compound (A) is:

(1) 

(2) 

(3) 

(4) 

90. At 25 °C, the molar conductances at infinite dilution for the strong electrolytes NaOH, NaCl and $BaCl_2$ are 248×10^{-4}, 126×10^{-4} and 280×10^{-4} $Sm^2\ mol^{-1}$ respectively, λ°_m $Ba(OH)_2$ in $Sm^2\ mol^{-1}$ is:

(1) 52.4×10^{-4}

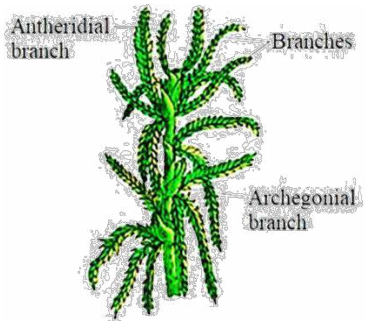
(2) 524×10^{-4}

(3) 402×10^{-4}

(4) 262×10^{-4}

[Contd...

91. Identify the **correct** statement for the plant shown in the given diagram.



(1) It produces naked seeds.
(2) It is the *Sphagnum* gametophyte.
(3) It is the *Sphagnum* sporophyte.
(4) The seeds are enclosed in fruits.

92. Identify the **correct** statement regarding the Z scheme of light reaction of photosynthesis.
(1) It occurs in the stroma lamellae.
(2) The characteristic shape is formed when all electron carriers are placed in a sequence on a light intensity scale.
(3) It starts with PS I, moves uphill to an acceptor, then downhill to PS II, and finally to $NADP^+$.
(4) It leads to production of ATP and NADPH + H^+.

93. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): The logistic growth model is considered a more realistic one for most animal populations.

Reason (R): Resources for growth for most animal populations are finite and become limiting sooner or later.

In the light of the above statements, choose the *most appropriate* answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

94. Match **List-I** with **List-II**.

List-I (Kingdom)		**List-II (Characteristic)**	
(A)	Monera	(I)	Tissue/organ/organ system body organisation
(B)	Protista	(II)	Non-cellulosic (Polysaccharide + amino acid) cell wall
(C)	Fungi	(III)	Eukaryotic cell type with presence of cell wall in some
(D)	Animalia	(IV)	Presence of cell wall with chitin

Choose the *most appropriate* answer from the options given below.
(1) A-II, B-III, C-IV, D-I
(2) A-II, B-I, C-IV, D-III
(3) A-III, B-II, C-I, D-IV
(4) A-IV, B-III, C-I, D-II

95. Consider the following statements about meiosis and select the option with the **correct** statements.
A. It occurs during gametogenesis in plants and animals.
B. Meiosis I is initiated before the parental chromosomes have replicated.
C. It involves pairing of non-homologous chromosomes.
D. Recombination occurs between sister chromatids of homologous chromosomes.
E. Four diploid cells are formed at the end of meiosis II.
(1) A only
(2) A and C only
(3) B, D and E only
(4) A, C and D only

96. The 1:2:1 phenotypic ratio in F_2 generation of dog flower plants for the inheritance of flower colour is due to;
(1) pleiotropy.
(2) incomplete dominance.
(3) multiple alleles.
(4) co-dominance.

97. A researcher is studying the decomposition rates of two different types of detritus in two different experimental setups:
Detritus A: Rich in lignin and chitin.
Detritus B: Rich in nitrogen and water-soluble sugars.
Setup I: The detritus is kept in a warm, moist environment with high aeration.
Setup II: The detritus is kept in a low-temperature environment with anaerobic conditions.
Based on the principles of decomposition, which of the following statements is most likely to be **correct**?
(1) The rate of decomposition for Detritus A in Setup I will be faster than Detritus B in Setup II, because environmental factors always override chemical composition.
(2) Detritus B in Setup I will show the highest rate of mineralization because both the chemical quality of the detritus and the climatic conditions are synergistic for microbial activity.
(3) Humification will occur faster in Setup II for Detritus B than in Setup I for Detritus A, because low temperatures promote the accumulation of dark amorphous substances.
(4) In Setup II, the rate of catabolism will be higher than the rate of fragmentation because bacterial enzymes are less affected by oxygen availability than detritivores like earthworms.

98. The meiotic spindle of prophase I is assembled during:
(1) Pachytene. (2) Diplotene.
(3) Diakinesis. (4) Zygotene.

99. Arrange the following techniques/processes in the correct order as they are used in DNA fingerprinting.
A. Autoradiography
B. Electrophoresis
C. DNA isolation
D. Hybridisation with VNTR probe
E. Southern blotting
Choose the **correct** option.
(1) B → A → D → C → E
(2) B → C → A → D → E
(3) A → B → C → D → E
(4) C → B → E → D → A

100. Identify the **incorrectly** matched pair.
(1) Phycomycetes - *Mucor*
(2) Ascomycetes - *Penicillium*
(3) Basidiomycetes - *Puccinia*
(4) Deuteromycetes - *Agaricus*

101. Due to a mutation, a yeast cell produces defective RNA polymerase III. What will be its most likely consequence?
(1) Synthesis of 5.8S rRNA in it will stop.
(2) The catalytic RNA will not be produced inside its mitochondria.
(3) Polypeptide synthesis in cytoplasm will not take place.
(4) Polynucleotide synthesis in nucleus will be compromised.

102. Match **List-I** with **List-II** w.r.t chemiosmosis in chloroplast.

List-I		List-II	
(A)	Water-splitting complex	(I)	Present on the stroma side of thylakoid membrane
(B)	CF_1 of ATP synthase	(II)	Embedded in the thylakoid membrane
(C)	CF_0 of ATP synthase	(III)	Present on the inner side of thylakoid membrane

Choose the *most appropriate* answer from the options given below.
(1) A-III, B-II, C-I (2) A-III, B-I, C-II
(3) A-I, B-III, C-II (4) A-I, B-II, C-III

103. In Morgan's experiments on *Drosophila*, the genes for white eye and yellow body showed ____ recombination.
(1) 98.7% (2) 37.2%
(3) 62.8% (4) 1.3%

104. Identify the group of organisms that are unicellular, colonial, or filamentous, and possess chlorophyll *a* and *b* localized in definite chloroplasts.
(1) Phaeophyceae (2) Rhodophyceae
(3) Chlorophyceae (4) Cyanobacteria

105. The growth of pollen tube is measured in terms of increase in:
(1) length. (2) cell size.
(3) cell number. (4) surface area.

[Contd...

106. The main function of a particular organelle in a eukaryotic cell is packaging of materials. Choose the **incorrect** statement about it.
(1) It is a part of the endomembrane system.
(2) Its functions are not coordinated with that of endoplasmic reticulum.
(3) It is bound by a single membrane.
(4) It has two distinct faces viz. *cis* and *trans*.

107. Given below are two statements:
Statement I: Ecology is one of the threads which gives us a holistic perspective to biology.
Statement II: A particular aspect of ecology is the study of anthropogenic environmental degradation and the socio-political issues it has raised.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

108. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): In prokaryotes, translation can begin much before the mRNA is fully transcribed.
Reason (R): In bacteria, splicing of the primary transcript begins as soon as it is synthesised.
In the light of the above statements, choose the *most appropriate* answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

109. In an experimental setup, four sets of same plant species are treated as follows:
Set P: Shoot tip is removed (Decapitation).
Set Q: Shoot tip is intact, but a high concentration of Cytokinin is applied directly to the lateral buds.
Set R: Shoot tip is removed, and an agar block containing Abscisic Acid (ABA) is placed on the cut stump.
Set S: Shoot tip is intact, and the plant is kept in a water-stressed environment.
Based on the physiological effects of PGRs, in which sets would you expect to see the promotion of lateral bud growth (bushy growth)?
(1) P and Q only (2) P, Q, and R
(3) Q and S only (4) P and S only

110. The compound phosphoenol pyruvate is;
(1) formed from pyruvate during glycolysis.
(2) the primary acceptor of CO_2 in maize.
(3) a 5-carbon compound.
(4) withdrawn from the respiratory pathway for fatty acid synthesis.

111. The curd is rich in;
(1) vitamin B_{12}. (2) vitamin B_1.
(3) vitamin K. (4) vitamin B_5.

112. What percentage of progeny born to parents having aaBBCc and AabbCC genotypes for certain traits will have the genotype AabbCc (considering independent segregation of all alleles)?
(1) 25 (2) 75
(3) 0 (4) 50

113. The bottom strata of a forest ecosystem is occupied by;
(1) trees and shrubs.
(2) herbs and shrubs.
(3) grasses and trees.
(4) grasses and herbs.

114. During lactic acid fermentation, which molecule is oxidised?
(1) $NADP^+$ (2) $NADH + H^+$
(3) NAD^+ (4) $NADPH + H^+$

115. Match **List-I** with **List-II**.

	List-I		List-II
(A)	*Oscillatoria*	(I)	Yeast used in making bread and beer
(B)	*Nucleopolyhedrovirus*	(II)	Biofertiliser
(C)	*Saccharomyces cerevisiae*	(III)	Yeast, source of statins
(D)	*Monascus purpureus*	(IV)	Bicontrol agent

Choose the *most appropriate* answer from the options given below.
(1) A-II, B-IV, C-I, D-III
(2) A-IV, B-II, C-III, D-I
(3) A-II, B-I, C-IV, D-III
(4) A-III, B-IV, C-I, D-II

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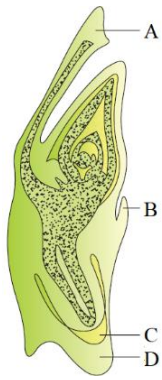
116. In the fruit of coconut, the mesocarp is:
(1) edible. (2) fleshy.
(3) stony and hard. (4) fibrous.

117. Amongst the following places, the highest number of bird species is found in;
(1) Colombia. (2) India.
(3) New York. (4) Greenland.

118. Given below are two statements:
Statement I: The '*I*' gene of ABO blood group system in humans is a pleiotropic gene.
Statement II: A pleiotropic gene can exhibit multiple phenotypic expression.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

119. Identify the **incorrect** statement regarding the anatomy of a dicotyledonous root.
(1) The outermost layer is the epiblema.
(2) The pith is large and well developed.
(3) The endodermis comprises a single layer of barrel-shaped cells.
(4) All tissues on the innerside of the endodermis, such as pericycle, vascular bundles, and pith, constitute the stele.

120. Which labelled structure in the following diagram of L.S. of an embryo of grass is a rudimentary cotyledon?



(1) A (2) B
(3) C (4) D

121. Consider the following statements regarding population interactions and select the option with all **correct** statements.
A. In predation, only one species benefits and the interaction is detrimental to the other.
B. In commensalism, one species is benefitted and the other is neither benefitted nor harmed.
C. In amensalism, both species are harmed.
D. In competition, both the interacting species lose.
E. In mutualism, one species is harmed while the other is unaffected.
(1) B, C and D (2) A, C and E
(3) A, B and D (4) A, D and E

122. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): Pteridophytes lack vascular tissues – xylem and phloem.
Reason (R): In pteridophytes, the main plant body is a sporophyte which is differentiated into true root, stem and leaves.
In the light of the above statements, choose the *most appropriate* answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

123. A student while studying cell cycle observes two sets of chromosomes and all chromosomes have their centromeres directed towards the poles. Most likely, the cell is in;
(1) Prophase. (2) Metaphase.
(3) Anaphase. (4) Telophase.

124. For photosynthesis, there is a _____ relationship between incident light and CO_2 fixation rates at low light intensities.
(1) inverse (2) linear
(3) hyperbolic (4) parabolic

125. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Frederick Griffith	(I)	Proved DNA is the genetic material using bacteriophages
(B)	Hershey and Chase	(II)	Proposed the 'Central dogma' of molecular biology
(C)	Watson and Crick	(III)	Proposed the double helix model for DNA structure
(D)	Francis Crick	(IV)	Discovered the 'Transforming Principle' in *Streptococcus pneumoniae*

Choose the *most appropriate* answer from the options given below.

(1) A-IV, B-I, C-III, D-II
(2) A-IV, B-III, C-I, D-II
(3) A-I, B-IV, C-II, D-III
(4) A-III, B-I, C-IV, D-II

126. Which of the following cells/tissues in plants are parenchymatous?

A. Pith
B. Hypodermis in stem
C. Mesophyll
D. Fruit wall of nuts
E. Companion cells

Select the **correct** option.

(1) A and D only (2) B and C only
(3) A, C and E only (4) A, B and D only

127. The malarial parasite:

(1) is multicellular.
(2) has cilia for locomotion.
(3) is a sporozoan.
(4) possesses gullet.

128. A cell organelle that is a membrane-bound space containing water, sap, and excretory products, and is bounded by a tonoplast, is;

(1) lysosome.
(2) vacuole.
(3) endoplasmic reticulum.
(4) golgi apparatus.

129. The 'rivet popper hypothesis' to explain the importance of species diversity in an ecosystem was proposed by:

(1) David Tilman.
(2) Alexander von Humboldt
(3) Edward Wilson.
(4) Paul Ehrlich.

130. Which of following organisms are placed in same 'order'?

(1) Monkey and Gibbon
(2) Wheat and Mango
(3) *Petunia* and *Triticum*
(4) Dog and Human

131. A pollen mother cell in a typical angiosperm;

(1) is haploid.
(2) undergoes microsporogenesis.
(3) gives rise to precursor of female gametophyte.
(4) develops in a megasporangium.

132. All the following possess cell walls, **except**;

(1) bladderwort. (2) tobacco mosaic virus.
(3) *Gonyaulax*. (4) spores of slime moulds.

133. The complex of mitochondrial ETS that does not have Fe-S centre is;

(1) complex I. (2) complex II.
(3) complex III. (4) complex IV.

134. Identify the **correct** statement.

(1) Intercalary meristem appears early in life of a plant.
(2) Intercalary meristem contributes to the formation of the secondary plant body.
(3) Apical meristems are also called cylindrical meristems.
(4) In grasses, fascicular vascular cambium regenerates parts removed by the grazing herbivores.

135. Select the option which represents the correct floral formula for a flower from the family Leguminosae.

(1) $\oplus\ K_{(5)}\ C_{(5)}\ A_5\ \underline{G}_{(2)}$
(2) $\%\ K_{(5)}\ C_{1+2+(2)}\ A_{(9)+1}\ \underline{G}_1$
(3) $Br\ \oplus\ P_{(3+3)}\ A_{3+3}\ \underline{G}_{(3)}$
(4) $\%\ K_{(5)}\ C_{(5)}\ A_{(5)}\ \underline{G}_{(2)}$

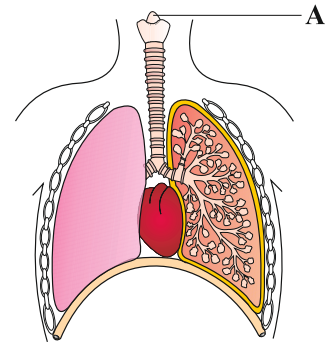
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136. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.
Assertion A: In ctenophores, the body bears eight external rows of ciliated comb plates, which are completely absent in cnidarians.
Reason R: Ctenophores exhibit organ-system level of organisation and are triploblastic, showing a major evolutionary advancement over diploblastic cnidarians.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both A and R are correct and R is the correct explanation for A.
(2) Both A and R are correct but R is not the correct explanation for A.
(3) A is correct but R is incorrect.
(4) A is incorrect but R is correct.

137. Read the following statements.
A. There is a wide diversity in living organisms in our biosphere.
B. Few elements present in a sample of earth's crust are present in a sample of living tissue.
C. All the carbon compounds that we get from living tissues can be called biomolecules.
D. Co-enzymes associate with the apoenzyme transiently, usually during catalysis.
E. Adenosine, guanosine, and uridine are examples of nucleotides found in nucleic acids.
Select the **correct** combination of statements:
(1) A, B, and D only
(2) B, C, and E only
(3) A, C, and D only
(4) A, B, D, and E only

138. Which of the following accurately describes the boundaries of genital pouch in male cockroaches?
(1) Dorsally by 8th and 9th terga and ventrally by 8th sternum
(2) Dorsally by 9th and 10th terga and ventrally by the 9th sternum
(3) Dorsally by 9th and 10th sterna and ventrally by the 9th tergum
(4) Dorsally by 7th and 8th terga and ventrally by the 7th sternum

139. Consider the diagrammatic view of the human respiratory system.



If a functional impairment occurs specifically at the level of the '**A**', which of the following processes will be most directly and immediately compromised?
(1) The diffusion of gases across the alveolar membrane.
(2) The generation of sound by the cartilaginous box.
(3) The prevention of food entry into the larynx during swallowing.
(4) The conduction of atmospheric air to the terminal bronchioles.

140. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Neutrophils	(I)	20-25 percent
(B)	Basophils	(II)	60-65 percent
(C)	Lymphocytes	(III)	6-8 percent
(D)	Monocytes	(IV)	0.5-1 percent

Choose the **correct** answer from the options given below:
(1) A-II, B-IV, C-I, D-III
(2) A-II, B-I, C-IV, D-III
(3) A-III, B-IV, C-I, D-II
(4) A-I, B-IV, C-II, D-III

141. Which of the following events acts as a direct check on the Renin-Angiotensin mechanism?
(1) Conversion of Angiotensin I to Angiotensin II.
(2) Secretion of Aldosterone.
(3) Release of ANF from the atria of the heart.
(4) Increased release of antidiuretic hormone.

142. Which of the following proteins plays a direct role in muscle contraction?
(1) Troponin
(2) Insulin
(3) Myoglobin
(4) Trypsin

[Contd...

143. The phase of repolarisation is dependent on which of the following across the axonal membrane?
(1) Rapid influx of Na^+ ions into the axoplasm.
(2) Rapid efflux of K^+ ions outside the membrane.
(3) Active transport of 3 Na^+ outwards and 2 K^+ inwards by the sodium-potassium pump.
(4) Massive influx of negatively charged proteins into the axoplasm.

144. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Melatonin	(I)	Juxtaglomerular cells of kidney
(B)	Thymosin	(II)	Pineal gland
(C)	Erythropoietin	(III)	Gastro-intestinal tract
(D)	Cholecystokinin	(IV)	Thymus gland

Choose the **correct** answer from the options given below:
(1) A-IV, B-II, C-I, D-III
(2) A-II, B-I, C-III, D-IV
(3) A-II, B-IV, C-I, D-III
(4) A-I, B-IV, C-II, D-III

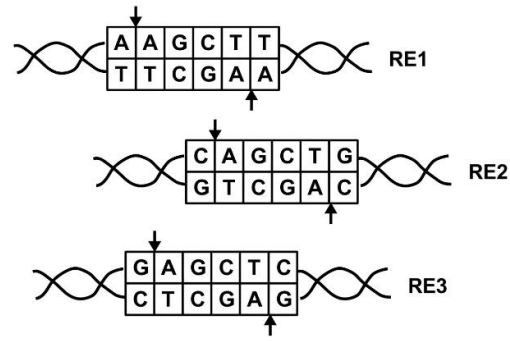
145. Given below are two statements:
Statement I: Soon after the infant is delivered, the placenta is also expelled out of the uterus.
Statement II: In human females, new oogonia continue to be formed and added steadily from birth until the onset of puberty.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

146. Which of the following works on principle of avoiding chances of the ovum and sperms meeting?
(1) Diaphragms and Vaults
(2) Progestasert and LNG-20
(3) Periodic abstinence and Coitus interruptus
(4) Multiload 375 and Lippes loop

147. The forelimbs of whales, bats, cheetahs, and humans is a classic example of:
(1) Analogy resulting from convergent evolution
(2) Homology resulting from divergent evolution
(3) Analogy resulting from divergent evolution
(4) Homology resulting from convergent evolution

148. Identify the **incorrectly** matched pair.
(1) *Plasmodium falciparum*: Responsible for malignant malaria which can be fatal.
(2) *Wuchereria bancrofti*: Causes chronic inflammation of lymphatic vessels
(3) *Microsporum*: Pathogenic bacterium responsible for severe pneumonia in humans.
(4) *Entamoeba histolytica*: Protozoan parasite causing amoebic dysentery

149. Provided below are recognition sequences for restriction enzymes 1 (RE1), 2 (RE2) and 3 (RE3). Arrows indicate the positions where the enzymes digest on the two strands. Which of the following can the RE1 digested DNA ligate to?



(1) All three RE1, RE2 and RE3 digested DNA.
(2) Only to RE1 digested DNA.
(3) Only to RE1 and RE3 digested DNA.
(4) Only to RE1 and RE2 digested DNA.

150. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.
Assertion A: Transgenic mice are being developed for use in testing the safety of vaccines.
Reason R: Transgenic animals carry genes which make them resistant to toxic substances, ensuring they do not die during initial chemical safety testing.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

151. An aquatic vertebrate has to swim constantly to avoid sinking. This organism belongs to the class:
(1) Osteichthyes
(2) Cyclostomata
(3) Chondrichthyes
(4) Amphibia

152. In an exothermic reaction, how does the energy level of the product (P) relate to the substrate (S)?
(1) 'P' is at a higher energy level than 'S'.
(2) 'P' is at a lower energy level than 'S'.
(3) 'P' and 'S' are at the exact same energy level.
(4) The energy of 'P' depends entirely on the presence of the competitive inhibitor.

153. Identify the **incorrect** statements regarding the respiratory system of a frog.
A. During aestivation and hibernation, gaseous exchange takes place through the lungs.
B. In water, the skin acts as an aquatic respiratory organ via diffusion.
C. The lungs are a pair of elongated, pink-colored sac-like structures found in the abdominal cavity.
D. Buccal cavity respiration is the primary mode of exchange when the frog is submerged in water.
(1) A, B, C, and D only
(2) B, C and D only
(3) A, C, and D only
(4) A, B, and D only

154. Which of the following capacities or volumes **cannot** be directly estimated using spirometry?
(1) IC (2) ERV
(3) VC (4) FRC

155. If a patient, in his ECG, have exactly 80 QRS complexes in one minute, and a stroke volume of 70 mL, what is his approximate cardiac output?
(1) 5000 mL/min (2) 5600 mL/min
(3) 4800 mL/min (4) 6200 mL/min

156. On an average, _____ml of blood is filtered by the kidneys per minute which constitute roughly _____ of the blood pumped out by each ventricle of the heart in a minute
(1) 500-600, $1/5^{th}$ (2) 1100-1200, $l/3^{rd}$
(3) 500-600, $l/3^{rd}$ (4) 1100-1200, $l/5^{th}$

157. In the human axial skeleton, the term bicephalic is used to describe the ribs because:
(1) They articulate with both the sternum ventrally and the clavicle dorsally.
(2) They possess two articulation surfaces on their dorsal end
(3) They are formed by the fusion of two separate cartilaginous embryonic precursors.
(4) They have two distinct heads that articulate solely with the hyoid bone.

158. Anatomically, the limbic system is composed of**:**
(1) The midbrain, pons and medulla oblongata.
(2) The cerebral cortex, corpus callosum, thalamus.
(3) The inner parts of the cerebral hemispheres and structures like the amygdala and hippocampus.
(4) The corpora quadrigemina and the cerebral aqueduct.

159. Which of the following is hypercalcemic hormone?
(1) Thyrocalcitonin (TCT)
(2) Aldosterone
(3) Parathyroid hormone (PTH)
(4) Cortisol

160. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.
Assertion A: The mitotic division starts as the zygote moves through the isthmus of the oviduct called cleavage towards the uterus and forms 2, 4, 8, 16 daughter cells called blastomeres.
Reason R: The embryo with 8 to 16 blastomeres is called a blastula and it continues to divide and transforms into blastocyst.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

161. The primary medical and social rationale for statutory ban on amniocentesis was to:
(1) prevent the transfer of genetic disorders.
(2) ensure massive child immunisation protocols are followed post-delivery.
(3) legally check the increasing menace of indiscriminate female foeticide.
(4) promote the widespread use of 'Saheli'.

162. Given below are two statements:
Statement I: Oparin and Haldane proposed that the first form of life could have come from pre-existing non-living organic molecules.
Statement II: S.L. Miller's 1953 laboratory experiment successfully proved the theory of spontaneous generation.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

163. This exaggerated immune response to allergens is primarily mediated by which type of antibodies and involves the release of which chemicals?
(1) IgG antibodies; release of interferons from macrophages.
(2) IgA antibodies; release of colostrum from secretory glands.
(3) IgE antibodies; release of histamine and serotonin from mast cells.
(4) IgM antibodies; release of antitoxins from T-lymphocytes.

164. Match **List-I** with **List-II**.

List-I		List-II	
(A)	ori	(I)	Usually an antibiotic resistance gene used to identify transformants.
(B)	Selectable marker	(II)	Loss of an enzyme's activity due to the insertion of alien DNA.
(C)	Cloning site	(III)	Initiates replication and controlling copy number.
(D)	Insertional inactivation	(IV)	Specific recognition sequence where restriction enzymes cut to link alien DNA.

Choose the **correct** answer from the options given below:
(1) A-III, B-II, C-IV, D-I
(2) A-IV, B-II, C-III, D-I
(3) A-I, B-IV, C-II, D-III
(4) A-III, B-I, C-IV, D-II

165. Which specific polypeptide segment is removed during the maturation process of human insulin?
(1) Chain A (2) Chain B
(3) The C peptide (4) Disulphide bridges

166. Which of the following phyla predominantly consists of organisms showcasing bilateral symmetry along with a **true** coelom?
(1) Ctenophora
(2) Coelenterata
(3) Platyhelminthes
(4) Annelida

167. In a specific pH, the amino acid exists as a molecule carrying both a positive and a negative charge. This form is uniquely referred to as:
(1) Pro-hormone form
(2) Zwitterionic form
(3) Agranulocyte form
(4) Polypeptide form

168. If the head of a cockroach is severed, it can survive for several days. Why?
(1) The brain is highly consolidated in the thorax, allowing it to bypass the head entirely.
(2) The nervous system is represented by a single dorsal nerve cord that operates autonomously.
(3) The head holds only a tiny portion of the nervous system, while the rest is situated along the ventral part of its body.
(4) Malpighian tubules take over the sensory functions of the head after decapitation.

169. At the alveolar site, which condition drives the reaction towards the formation of CO_2 and H_2O from bicarbonate?
(1) High pCO_2 and high pO_2.
(2) Low pCO_2 and low H^+ concentration.
(3) High pO_2 and high pCO_2.
(4) High H^+ concentration and high temperature.

170. Select the **incorrectly** matched pair regarding disorders of the human circulatory system:
(1) Atherosclerosis: Deposition of calcium, fat, cholesterol, and fibrous tissue
(2) Angina pectoris: Acute chest pain, when not enough oxygen is reaching the heart muscle
(3) Heart Failure: The heart completely stops beating suddenly
(4) Hypertension: Blood pressure that is persistently higher than the normal 120/80 mm Hg.

171. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Micturition	(I)	Eliminate sterols
(B)	Diabetes mellitus	(II)	Nervous system control
(C)	Kidney failure	(III)	Glucose in urine
(D)	Sebaceous glands	(IV)	Hemodialysis

Choose the **correct** answer from the options given below:
(1) A-IV, B-III, C-I, D-II
(2) A-IV, B-I, C-III, D-II
(3) A-II, B-III, C-IV, D-I
(4) A-II, B-IV, C-III, D-I

172. Evaluate the following statements and identify them as **True (T)** or **False (F)**:
A. The 8th, 9th and 10th pairs of ribs articulate directly with the sternum via hyaline cartilage.
B. The joint between the carpal and metacarpal of the human thumb is a saddle joint.
C. Myasthenia gravis is an auto-immune disorder affecting the neuromuscular junction leading to fatigue and paralysis of skeletal muscle.
D. The acromion process is found on the clavicle.
(1) A-T, B-T, C-F, D-T
(2) A-F, B-T, C-T, D-F
(3) A-F, B-F, C-T, D-F
(4) A-T, B-F, C-T, D-T

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173. As opposed to DNA replication within the cell, discontinuous synthesis of DNA does not occur in a polymerase chain reaction (PCR). Why?

(1) Denaturation step in PCR substitutes for the replication fork and *Taq* polymerase extends DNA only in the 5′ to 3′ direction.

(2) The replication fork is formed and *Taq* polymerase extends DNA in both 3′ to 5′ and 5′ to 3′ direction.

(3) Denaturation step in PCR substitutes for the replication fork and *Taq* polymerase extends DNA only in the 3′ to 5′ direction.

(4) The replication fork is formed and DNA ligase activity of the *Taq* polymerase joins the discontinuous fragments.

174. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: TSH acts directly on body tissues to regulate metabolism.

Reason R: TSH stimulates thyroid gland to release thyroxine.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true but R is not the correct explanation of A.

(3) A is true but R is false.

(4) A is false but R is true.

175. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Menarche	(I)	Release of mature sperms into lumen of seminiferous tubules
(B)	Spermiation	(II)	The first menstruation beginning at puberty.
(C)	Colostrum	(III)	The process of childbirth
(D)	Parturition	(IV)	Milk produced during initial days of lactation.

Choose the **correct** answer from the options given below:

(1) A-III, B-I, C-IV, D-II

(2) A-I, B-II, C-III, D-IV

(3) A-II, B-IV, C-I, D-III

(4) A-II, B-I, C-IV, D-III

176. For which one of the following diseases does the causative agent require the splicing of their hnRNA to generate mature mRNA?

(1) Malaria (2) Pertussis

(3) Typhoid (4) Tuberculosis

177. In an island with 10,000 individuals, four have sickle cell anaemia, a recessive autosomal disease. Assuming that the locus is in Hardy-Weinberg equilibrium, how many individuals in that island are expected to be heterozygous for the disease allele?

(1) 392 (2) 4

(3) 9996 (4) 9608

178. Given below are two statements:

Statement I: During HIV infection, the virus enters macrophages where the viral RNA genome replicates using reverse transcriptase, effectively turning the macrophage into an HIV factory.

Statement II: The primary cause of the severe immuno-deficiency in an AIDS patient is the progressive decrease in the number of B-lymphocytes.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement I and Statement II are correct.

(2) Both Statement I and Statement II are incorrect.

(3) Statement I is correct but Statement II is incorrect.

(4) Statement I is incorrect but Statement II is correct.

179. All of the following are the characteristics of cloning vector pBR322, **except**;

(1) It has two antibiotic resistance genes.

(2) It is a chromosomal DNA.

(3) It has low copy number than the bacteriophage.

(4) It's size is about 4.3kb.

180. How does RNA interference method achieve parasite resistance in the transgenic host plant?

(1) It produces a toxic crystalline protein that lyses the midgut epithelial cells of the nematode.

(2) It introduces DNA that produces both sense and anti-sense RNA, forming a dsRNA

(3) It triggers the plant's natural immune system to secrete alpha-interferon against the nematode.

(4) It creates a mechanical barrier in the roots that the nematode cannot physically penetrate.

■■■

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